

ANASUYA ACHARYA

Postdoctoral Researcher at Aarhus University Interested in Cryptography

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PUBLICATIONS

Conference Proceedings

- Acharya, Anasuya, Karen Azari, Mirza Ahad Baig, Dennis Hofheinz, and Chethan Kamath (2025). "Securely Instantiating 'Half Gates' Garbling in the Standard Model". In: *PKC 2025, May 12-15, 2025*. Røros, Norway: Springer, pp. 37–75.
- Acharya, Anasuya, Karen Azari, and Chethan Kamath (2025). "On the Adaptive Security of Free-XOR-Based Garbling Schemes in the Plain Model". In: *EUROCRYPT 2025, May 4-8, 2025*. Madrid, Spain: Springer, pp. 214–244.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2025). "Towards Building Efficient SCALES Protocols". In: *ASIACRYPT 2025, December 8-12, 2025*. Melbourne, Australia: Springer.
- Acharya, Anasuya, Carmit Hazay, and Muthuramakrishnan Venkitasubramaniam (2025). "On Achieving 'Best-in-the-Multiverse' MPC". In: *TCC 2025, December 1-5, 2025*. Aarhus, Denmark: Springer.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2024). "Malicious Security for SCALES - Outsourced Computation with Ephemeral Servers". In: *CRYPTO 2024, August 18-22, 2024*. Santa Barbara, USA: Springer, pp. 3–38.
- Acharya, Anasuya, Tomer Ashur, Efrat Cohen, Carmit Hazay, and Avishay Yanai (2023). "A New Approach to Garbled Circuits". In: *ACNS 2023, June 19-22*. Kyoto, Japan: Springer, pp. 611–641.
- Acharya, Anasuya, Carmit Hazay, Oxana Poburinnaya, and Muthuramakrishnan Venkitasubramaniam (2023). "Best of Both Worlds - Revisiting the Spymasters Double Agent Problem". In: *CRYPTO 2023, August 20-24, 2023*. Santa Barbara, USA: Springer, pp. 328–359.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2022). "SCALES - MPC with Small Clients and Larger Ephemeral Servers". In: *TCC 2022, November 7-10, 2022*. Chicago, USA: Springer, pp. 502–531.
- Prabhakaran, Manoj M., Anasuya Acharya, and Akash Trehan (2019). "An Introduction to the CellTree Paradigm". In: *ICISS 2019, Volume 11952 LNCS, 16-20 December 2019*. Hyderabad, India: Springer.
- Sanyashi, Tikaram, Anasuya Acharya, and Bernard Menezes (2019). "Plaintext Recovery Attacks and their Mitigation in an Application Specific SHE Scheme". In: *PDCAT 2019, 5-7 December 2019*. Gold Coast, Australia: IEEE Xplore.
- Patil, Vishwas, Anasuya Acharya, and R. K. Shyamasundar (2018). "Landcoin: A Land Management System using Litecoin Blockchain Protocol". In: *SDLT3 2018, 12 November 2018*. Gold Coast, Australia.

REVIEWING

Subreviewer for Several IACR Conferences

CRYPTO '22, '24, '25
ITC '25

EUROCRYPT '22, '24, '25
ASIACRYPT '25

TCC '22, '24
PKC '26

EDUCATION

Ph.D. in Computer Engineering

Bar-Ilan University

July 2025

Ramat Gan, Israel

CGPA: 86.94 / 100

M.Tech. in Computer Science and Engineering (CSE)

IIT Bombay

July 2020

Mumbai, India

CGPA: 9.54 / 10

B.Tech. in CSE

NIT Trichy

May 2017

Tiruchirappalli, India

CGPA: 8.06 / 10

REFEREES

Prof. Claudio Orlandi

Aarhus University

Aarhus, Denmark

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Postdoc Host

Prof. Carmit Hazay

Bar-Ilan University

Ramat Gan, Israel

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PhD Thesis Supervisor

Prof. Manoj M. Prabhakaran

IIT Bombay

Mumbai, India

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www.cse.iitb.ac.in/~mp/

Masters Thesis Supervisor

SERVICE

Workshops and Talks

- Maliciously Secure SCALES Protocols (2024). Cryptography in the Blockchain Era (CIBE) 2024, 23-27 June 2024, Bertinoro, Italy.
- Maliciously Secure SCALES Protocols (2024). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2024, 3-6 June 2024, TU Darmstadt, Germany.
- SCALES: MPC with Small Clients and Larger Ephemeral Servers (2023). ACE Conference (Algorand Centres of Excellence) 2023, 10-12 January 2023, Hotel Arts, Barcelona.
- SCALES: MPC with Small Clients and Larger Ephemeral Servers (2022). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2022, 7-10 June 2022, Aarhus University, Denmark.
- CellTree: A new Paradigm for Distributed Data Repositories (2019). Talk, 28 June 2019, IIT Bombay, Mumbai, India.

Organizing

- 39th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (2019). 11-13 December 2019, IIT Bombay, Mumbai, India.
- ACM India Grad Cohort 2018 (2018). 6-7 July 2018, IIT Bombay, Mumbai, India.

EXPERIENCE

Research Assistant

Information Security Research and Development Center (ISRDC)

July 2017 – July 2020 Mumbai, India

A three year scholarship student and Research Assistant in ISRDC, IITB.

IT Intern

Atreus Global

May – June 2016 Jakarta, Indonesia

Java developer for SAP installation, integration and documentation.

Research Intern

Indian Statistical Institute

December 2015 – January 2016 Kolkata, India

Research intern in cryptography under Prof. Shubhamoy Maitra, studying and attempting cryptanalysis of the RC4 stream cipher.

Systems Engineering Intern

P.T. Yokogawa Indonesia

May – June 2015 Jakarta, Indonesia

Involved in installation project of DCS (Distributed Control Systems) in industrial production plants.

IT Intern

P.T. Indorama Synthetics

December 2014 – January 2015 Purwakarta, Indonesia

Developer, tester and debugger for PHP based applications, and performed data analysis using Oracle BI.

ACHIEVEMENTS



GATE (Graduate Aptitude Test for Engineering)

Score of 791 and AIR (All India Rank) 231 in the GATE Computer Science and Information Technology paper in February 2017



AP (Advanced Placement Exam)

Scored 3/5 in Physics C Mechanics, Computer Science A, Calculus AB in May 2012



AAT (National Award of Academic Aptitude and Achievement Test)

Scored 99 percentile in quantitative aptitude in May 2007

SOFTWARE SKILL SET

LaTeX GIT Linux(Ubuntu)

Java C C++ SEBOL JavaScript

Python PHP MATLAB

LANGUAGES

English ●●●●●

Hindi ●●●●●

Bengali ●●●●●

Japanese ●●●●●

Mandarin ●●●●●

Bahasa Indonesia ●●●●●

EXTRA-CURRICULAR

JLPT N2 ●●●●●

HSK 3 ●●●●●

TEFL ●●●●●

BMC @ HMI ●●●●●

HOBBIES

Trekking Learning Languages

Writing Crash Coursing Visual Arts